

Khung code Gaussian Naive Bayes

```
import numpy as np
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split

class GaussianNaiveBayes:
    def fit(self, X, y):
        """
        Huan luyen mo hinh Naive Bayes.

        Tham so
        -----
        X : np.ndarray, shape (n_samples, n_features)
            Ma tran du lieu huan luyen
        y : np.ndarray, shape (n_samples,)
            Vector nhan cua du lieu huan luyen

        Ket qua can luu:
        - self.params[c]["mean"] : np.ndarray, shape (n_features,)
        - self.params[c]["var"]   : np.ndarray, shape (n_features,)
        - self.priors[c]          : float (xac suat tien nghiem cua lop
                                   c)
        """
        # TODO: viet code o day
        pass

    def gaussian_pdf(self, x, mean, var):
        """
        Tinh mat do Gaussian cho vector x.

        Tham so
        -----
        x : np.ndarray, shape (n_features,)
            Vector dac trung cua mot mau
        mean : np.ndarray, shape (n_features,)
            Vector trung binh theo tung dac trung cua lop
        var : np.ndarray, shape (n_features,)
            Vector phuong sai theo tung dac trung cua lop

        Tra ve
        -----
        pdf : np.ndarray, shape (n_features,)
            Xac suat  $P(x_i | \text{lop})$  cho tung dac trung
        """
        # TODO: viet code o day
```

```

        pass

def predict_single(self, x):
    """
    Du doan nhan cho 1 mau.

    Tham so
    -----
    x : np.ndarray, shape (n_features,)
        Vector dac trung cua mot mau

    Tra ve
    -----
    label : int
        Nhan du doan (mot gia tri trong self.classes)
    """
    # TODO: viet code o day
    pass

def predict(self, X):
    """
    Du doan nhan cho nhieu mau.

    Tham so
    -----
    X : np.ndarray, shape (n_samples, n_features)
        Ma tran du lieu test

    Tra ve
    -----
    y_pred : np.ndarray, shape (n_samples,)
        Vector nhan du doan cho tung mau
    """
    # TODO: viet code o day
    pass

# -----
# PHAN MAIN (chay thu nghiem)
# -----
if __name__ == "__main__":
    # 1. Load du lieu Iris
    iris = load_iris()
    X, y = iris.data, iris.target # X: (150, 4), y: (150,)

    # 2. Chia train/test
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )
    # X_train: (120, 4), y_train: (120,)
    # X_test : (30, 4), y_test : (30,)

    # 3. Khoi tao va huan luyen model
    model = GaussianNaiveBayes()

```

```

model.fit(X_train, y_train)

# 4. Dự đoán
y_pred = model.predict(X_test) # y_pred: (30,)

# 5. Đánh giá
acc = np.mean(y_pred == y_test)
print("Accuracy:", acc)
print("Predicted:", y_pred)
print("True      :", y_test)

```

Tóm tắt shape cho sinh viên

- X_train: (n_samples, n_features) → với Iris là (120, 4)
- y_train: (n_samples,) → với Iris là (120,)
- mean: (n_features,) → với Iris là (4,)
- var: (n_features,) → (4,)
- x (một mẫu): (n_features,) → (4,)
- pdf (từng đặc trưng): (n_features,) → (4,)
- y_pred: (n_samples,) → (30,)